

FVC & onset site as predictors of ALSFRS-R decline in ALS

PRO-ACT cohort — bulbar declines faster and low baseline FVC predicts steeper decline, both small effects

PRO-ACT (Pooled Resource Open-Access ALS Clinical Trials), de-identified

n = 5,394 subjects with a fittable ALSFRS-R slope (0-18 months)

Headline: bulbar -0.23 pts/mo faster ($d = -0.25$); FVC% vs slope $r = 0.21$

Status: both hypotheses supported, effect sizes small, observational

Reproducibility: scripts/analysis.py, <1 min end-to-end

BOTH HYPOTHESES SUPPORTED — BUT EFFECT SIZES ARE SMALL; REPORT THEM AS SUCH

In **5,394** PRO-ACT patients with a fittable **ALSFRS-R** trajectory over their first 18 months, **bulbar-onset** disease declines faster than **limb-onset** (-1.24 vs -1.01 points/month), and **lower baseline forced vital capacity (FVC, % predicted)** predicts a steeper subsequent decline. Both associations are highly significant given the large sample, yet both are modest in magnitude (Cohen's $d = -0.25$; Pearson $r = 0.21$) and the individual distributions overlap heavily. These are real, reproducible cohort patterns — not deterministic rules for an individual patient.

**-0.23
pts/mo**

Extra ALSFRS-R loss per month, bulbar vs limb

Mixed-model group*time interaction; 95% CI $[-0.29, -0.16]$; $p = 6 \times 10^{-13}$; $n = 4,157$.

$d = -0.25$

Effect size, bulbar vs limb decline slope

"Small" by Cohen's convention; 95% CI $[-0.33, -0.18]$. Significant != large.

$r = 0.21$

Baseline FVC% vs decline slope

Weak positive; $p = 3 \times 10^{-18}$, $n = 1,766$. Higher FVC -> slower decline.

Hypotheses — formal statement & verdict

#	Claim (formal)	Status
H1	Mean ALSFRS-R decline slope differs by onset site (bulbar != limb)	Supported — small effect
H2	Baseline FVC (% predicted) is inversely associated with decline slope	Supported — weak effect
H3	The FVC->slope association survives adjustment for age, sex, disease duration, onset, riluzole	Supported (β essentially unchanged)

H1 (formal): $H0: \mu(\text{slope, bulbar}) = \mu(\text{slope, limb})$ vs $H1: \mu(\text{slope, bulbar}) \neq \mu(\text{slope, limb})$, where *slope* is the per-subject OLS decline rate of **ALSFRS-R** (points/month) over months 0–18.

H2/H3 (formal): in the model $\text{slope} = \beta_0 + \beta_1(\text{FVC}\%) + \gamma(z) + e$ (with e the error term), test $H0: \beta_1 = 0$ — first with covariates $z = \text{none}$ (unadjusted), then with $z = \{\text{age, sex, disease duration, onset, riluzole}\}$.

Data sources & pipeline

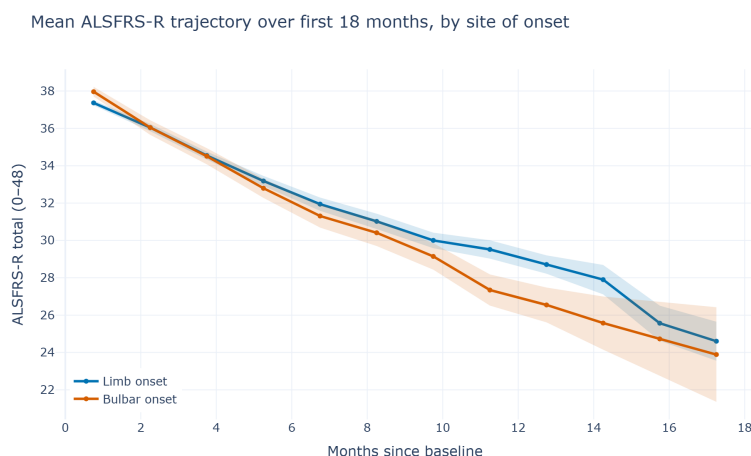
Data — PRO-ACT (Pooled Resource Open-Access ALS Clinical Trials), de-identified

- **F_PROACT_ALSFRS** — longitudinal **ALSFRS-R** total (0–48) with day offset *ALSFRS_Delta*; 51,947 scored visits.
- **F_PROACT_ALSHISTORY** — site of onset (bulbar / limb flags + text); onset date for disease duration.
- **F_PROACT_FVC** — spirometry, *pct_of_Normal* (FVC % predicted), best of ≤ 3 trials per visit.
- **F_PROACT_DEMOGRAPHICS** / **F_PROACT_RILUZOLE** — age, sex, riluzole use (confounders).
- Join key: **subject_id**. All reads from *data/PROACT_ALL_FORMS/*; nothing copied out.

Pipeline — one end-to-end script, ``scripts/analysis.py``

- 1 Fit each subject's ALSFRS-R slope over months 0–18 (≥ 3 visits, ≥ 90 -day span; median 8 visits/subject).
- 2 Assign pure-Bulbar vs pure-Limb onset (subjects flagged both / neither excluded).
- 3 Take baseline FVC% (earliest visit, window –30 to +90 days).
- 4 Welch t , Mann–Whitney, Cohen's d (CI), linear mixed-effects group*time; OLS slope~FVC%, unadjusted + adjusted.

Finding F1 — Bulbar onset declines faster: -1.24 vs -1.01 ALSFRS-R points/month



Mean ALSFRS-R trajectory over the first 18 months, bulbar vs limb onset, with 95% confidence bands.

SUPPORTED BY A MIXED-EFFECTS GROUP $\times$ TIME INTERACTION, $P = 6 \times 10^{-13}$

The two mean trajectories start together ($\sim 38/48$) and separate progressively: the bulbar curve sits below the limb curve from $\sim$ month 6 onward. The group $\times$ time interaction is -0.23 points/month (95% CI $[-0.29, -0.16]$) — bulbar patients lose roughly **4 extra ALSFRS-R points over 18 months**.

CLINICAL / MEDICAL

Bulbar-onset ALS begins in the brainstem motor nuclei governing speech and swallowing, and carries a worse prognosis than limb onset in most cohorts — earlier respiratory and nutritional compromise accelerate global functional loss captured by **ALSFRS-R**. The faster total-score decline here is consistent with that clinical picture.

DATA-SCIENCE / STATISTICS

Per-subject OLS slopes: bulbar mean -1.244 (SD 0.97, $n=930$), limb -1.012 (SD 0.90, $n=3,227$). Welch $t = -6.51$, $p = 1.0 \times 10^{-10}$; Mann-Whitney $p = 3.7 \times 10^{-13}$. The mixed model (random intercept + slope per subject) confirms the difference isn't an artifact of unequal visit counts.

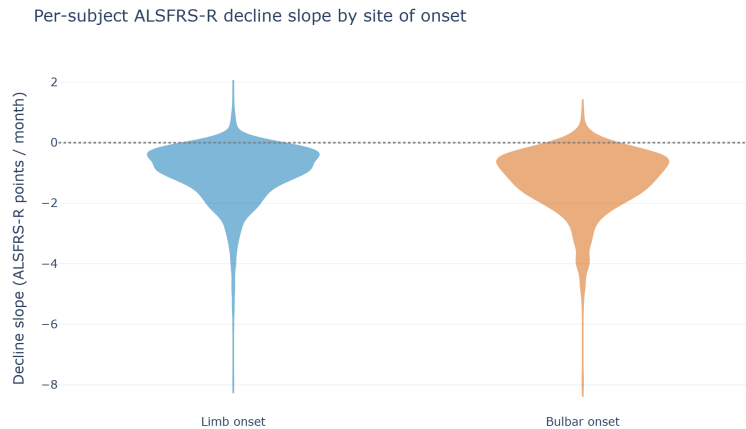
PLAIN ENGLISH

Think of two groups coasting downhill. Both start at the same height and both roll down, but the bulbar group's hill is slightly steeper, so after a year and a half they've dropped noticeably lower. On average — any single person can roll faster or slower than their group.

CONCRETE DATA EXAMPLE

Bulbar exemplar *subject_id* = 4453116: baseline ALSFRS-R 33, 14 visits, slope -1.065 /mo. Limb exemplar *subject_id* = 9971365: baseline 36, 7 visits, slope -0.833 /mo — a slower drop, illustrating the group tendency at the individual level.

Finding F2 — The difference is statistically robust but small: distributions overlap heavily



Per-subject decline-slope distributions by onset site (violin + box); dotted line marks zero/no-decline.

REAL BUT MODEST — COHEN'S D = -0.25 (SMALL); DO NOT OVER-INTERPRET

The group means differ, but the two slope distributions sit almost on top of each other. **Cohen's d = -0.25** (95% CI [-0.33, -0.18]) is a *small* effect: onset site shifts the average decline rate a little, and explains only a sliver of the person-to-person variation in how fast ALS progresses.

CLINICAL / MEDICAL

Clinically, this cautions against using onset site alone to prognosticate an individual. Many limb-onset patients decline faster than the average bulbar patient and vice versa; onset site is one weak axis among many (age, respiratory status, rate of spread).

DATA-SCIENCE / STATISTICS

With $n \sim 4,000$, a tiny mean gap (0.23 pts/mo) yields $p \sim 10^{-13}$ — a textbook case of the CLAUDE.md rule that in large cohorts significance $\neq$ importance. The honest summary is the effect size and its CI, not the p-value. Both non-parametric (Mann-Whitney) and parametric tests agree, so the *direction* is solid.

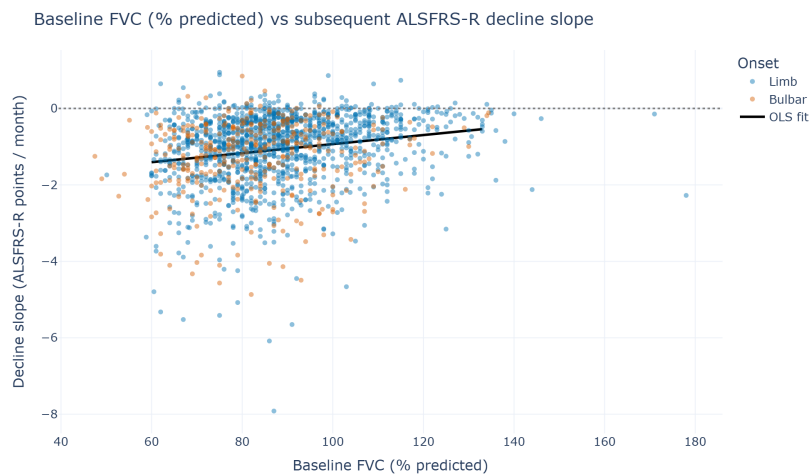
PLAIN ENGLISH

Two overlapping crowds of runners: the bulbar crowd is *on average* a touch slower to the finish, but if you pick one runner from each crowd you often can't tell which is which. The average difference is real; the overlap is huge.

CONCRETE DATA EXAMPLE

Bulbar median slope -1.066/mo vs limb median -0.833/mo — a 0.23-point/month median gap, mirroring the mean gap. The SDs (~0.9–0.97) are ~4x that gap, which is exactly why the violins overlap.

Finding F3 — Lower baseline FVC predicts faster decline, and it holds after adjustment



Baseline FVC (% predicted) vs subsequent ALSFRS-R decline slope, colored by onset; black line is the OLS fit.

SUPPORTED AND CONFOUNDER-ROBUST — ADJUSTED EFFECT UNCHANGED, $P = 3E-14$

Patients with lower baseline **FVC%** decline faster (more negative slope): Pearson $r = 0.21$ (95% CI [0.16, 0.25]), $n = 1,766$. Each 10-point drop in baseline FVC% predicts $\sim$ **0.13 points/month faster** ALSFRS-R decline. Adjusting for age, sex, disease duration, onset site and riluzole leaves the effect essentially unchanged (β 0.0118 $\rightarrow$ 0.0135).

CLINICAL / MEDICAL

FVC indexes respiratory (diaphragm/accessory) muscle strength, the system whose failure drives ALS mortality. A low baseline FVC signals disease that has already reached the respiratory motor pool, plausibly marking a more aggressive or more advanced phenotype — hence the faster global functional decline.

DATA-SCIENCE / STATISTICS

Unadjusted OLS: $\beta = 0.0118$ pts/mo per FVC% (95% CI [0.0092, 0.0145]), $R^2 = 0.042$. Adjusted OLS ($n=1,150$): $\beta = 0.0135$ (95% CI [0.0100, 0.0169]), $p = 2.8 \times 10^{-14}$, $R^2 = 0.136$. The predictor is weak (R^2 small) but stable under adjustment — consistent with an independent, not confounded, association.

PLAIN ENGLISH

Baseline breathing capacity is like the tread left on a tire: less tread at the start tends to mean a bumpier, faster-worsening road ahead. It nudges the odds — it doesn't dictate the destination, since most of the variation is still unexplained.

CONCRETE DATA EXAMPLE

Exemplar *subject_id* = 7511360: baseline FVC 69% predicted (low end) with a subsequent slope of -2.32 pts/mo — more than double the cohort-average decline, illustrating the low-FVC / fast-decline corner of the scatter.

What couldn't be refuted, and the caveats

Both hypotheses survived their tests

- **H1** (bulbar > limb decline) survives Welch t , Mann–Whitney, and a mixed-effects model — direction is robust.
- **H2/H3** (low FVC -> faster decline) survives confounder adjustment with a stable β — the association is not explained by age, sex, disease duration, onset, or riluzole.

What these results are NOT

- **Not large.** Both effects are small ($d = -0.25$; $r = 0.21$). Neither is an individual-level predictor.
- **Not causal.** This is observational cohort data — it confirms patterns, it does not prove that bulbar physiology or low FVC *cause* faster decline (per CLAUDE.md rule 2).
- **Not survivorship-free.** Slopes are conditioned on ≥ 3 visits in 18 months; the fastest-progressing patients (early dropout/death) are under-represented, likely *attenuating* both effects.

Methodology-pitfall checklist (references/methodology_pitfalls.md)

Circularity: avoided — groups defined by onset/FVC, outcome is an independent slope. **Confounders:** adjusted (age, sex, disease duration, onset, riluzole). **Survivorship:** acknowledged (18-mo window, ≥ 3 -visit filter). **Multiple testing:** only 3 pre-specified hypotheses; no fishing. **Causal overreach:** explicitly hedged.

Open questions for external review

Q1

Is the ≥ 3 -visit / 18-month filter attenuating the true bulbar effect?

Fast progressors drop out earliest. A joint (survival + longitudinal) model might recover a larger bulbar–limb gap. Worth a sensitivity analysis with an inverse-probability-of-dropout weight.

Q2

Should FVC% and onset be combined into one prognostic model?

Each is weak alone; a multivariable slope model (FVC% + onset + baseline ALSFRS-R + age) might reach useful prognostic R^2 . Do you want that as a follow-up?

Reproducibility — runs end-to-end on a laptop

- **1 script** — `projects/manu/fvc-slope/scripts/analysis.py` (~300 lines), `venv + requirements.txt` pinned.
- **3 figures** (PNG + interactive HTML), **1 results.json**, **1** subject-level CSV (local only).
- Reads `data/PROACT_ALL_FORMS/`; runtime < 1 minute. Re-run: `python scripts/analysis.py`.